VISUAL ENGINEERING SYNTHESIS

CAS-OHM QUANTIFICATION

Measuring attention flow through Cognitive Junction Transistors, RPMs, cascades, and recursive subcircuits.

$$V_{drive} = I_{attention} \times R_{constraint}$$

MEASURE · DESIGN · DIAGNOSE · RECURSE

A working guide that distinguishes source-defined CAS/TKS structures from the new decimal, four-container, ASOM, and residual-diagnostic extensions developed in this session.

WORKING DRAFT V0.1
July 2026

HOW TO READ THIS GUIDE

Source definitions vs. working extensions

SOURCE-DEFINED

What comes directly from the uploaded manuals

- CAS-Ohm: $V_{drive} = I_{attention} \times R_{constraint}$.
- CJT terminals: Cause, Bias, Effect, with gain β and four operating regions.
- D/W/P: Desire, Wisdom, and Power as prerequisites.
- The 12 face-card states as developmental phases.
- RPM: scan D → W → P, recurse on the first unmet prerequisite.
- CJT cascades: effect becomes the next cause; stage gains multiply.

WORKING MODEL

What this document adds

- D ↔ V, W ↔ I, and P ↔ capability-derived resistance.
- Decimal scoring and a four-container Q4 system.
- Trend tags to distinguish rising from waning states.
- ASOM as a mnemonic overlay on CJT operating regions.
- Predicted-versus-observed residuals for hidden-constraint diagnosis.
- Equivalent-resistance rules for nested sub-CJT assemblies.

Honesty rule: the numbers in this guide are design scores, calibration values, or observed ratings. They are not direct voltmeter readings from a human brain or a language model. The engineering value comes from consistency, test points, and falsifiable comparisons.

Notation discipline

Spell out **CAS-Ohm**. Do not use Ω as its shorthand inside CAS notation, because Ω already names emergent outcomes in CAS recipes and bindings.

Key conceptual distinction

Cause is not Drive. Cause is the incoming Idea or trigger. Drive is the potential pushing activation through the path. The two may be correlated, but they occupy different roles.

Reading map

Pages 3-7	Pages 8-12	Pages 13-18
Ohm law, placement, calculation modes, decimal system, Q4 containers	D/W/P mapping, card anchors, hybrid notation, ASOM, single-CJT example	Cascades, nested sub-CJTs, RPM recursion, residual diagnostics, field protocol, reference sheet

FOUNDATION

The CAS-Ohm isomorphism

$$V_{drive} = I_{attention} \times R_{constraint}$$

V
DRIVE POTENTIAL

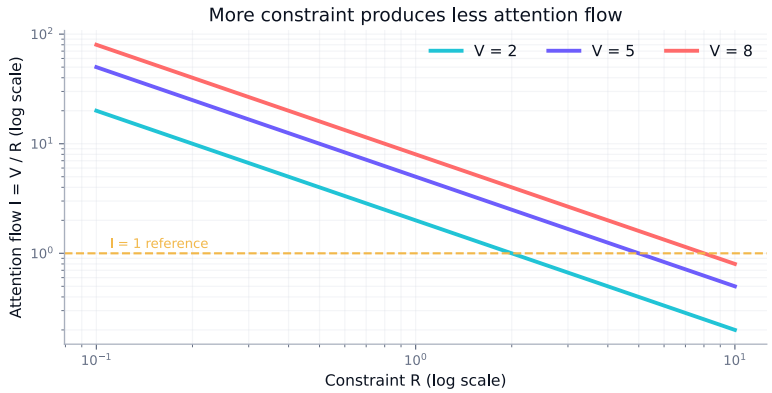
The motivational push or pressure behind a cognitive operation.

I
ATTENTION FLOW

The activation that actually moves through a cognitive path.

R
CONSTRAINT

Policies, capability limits, context, friction, and boundaries opposing flow.



At fixed Drive, raising constraint lowers attention flow. The relationship is proportional and testable.

Measure

$$I = V / R$$

Given Drive and constraint, calculate the attention flow.

Design

$$V = I \times R$$

Given a target flow and a known constraint, calculate the Drive required.

Diagnose

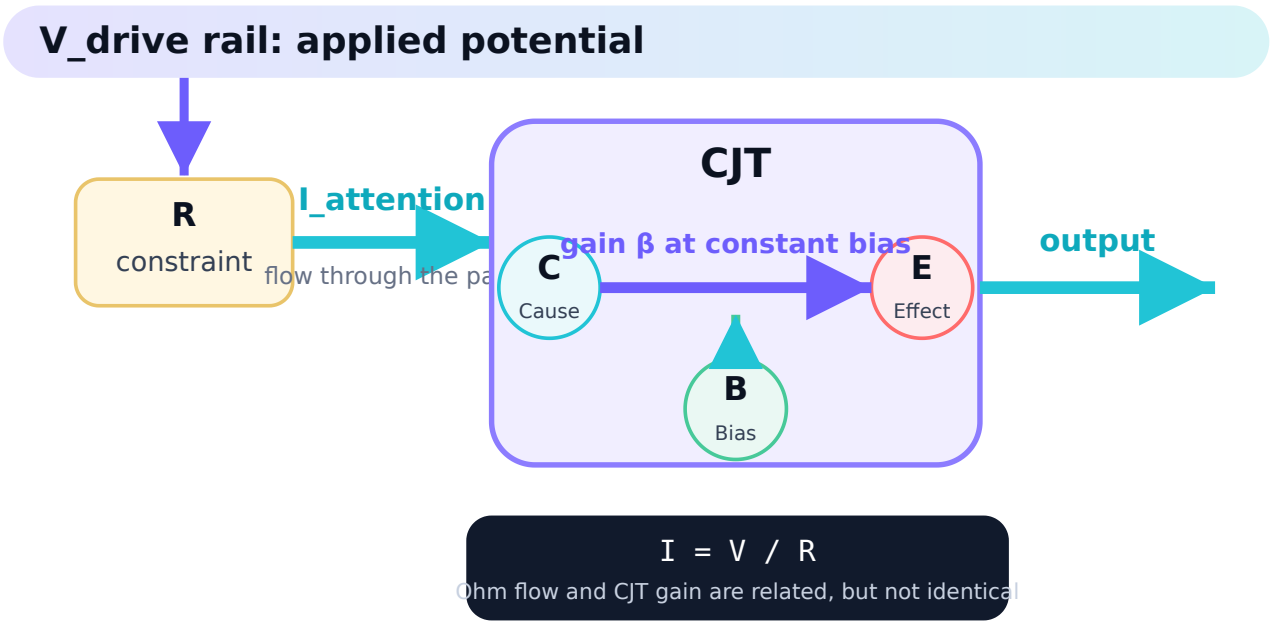
$$R = V / I$$

Given applied Drive and observed flow, infer the effective constraint.

SOURCE CAS PCBA Engineer, Chapter 5: the electrical and cognitive forms of Ohm's Law.

TOPOLOGY

Where Drive, attention, bias, and effect live



Proposed integrated view: a Drive rail applies potential, constraint limits path flow, and the CJT's bias controls how the cause becomes an effect.

Supply role

Calling V "outside" the CJT means it acts like a supply rail or applied potential. A CAS state can still carry a Drive tag, so outside does not mean disconnected from the state description.

Flow role

I is the attention or activation moving through a path. It is measured at a test point. The effect is the produced state, not the current itself.

Bias role

Bias is the accumulated substrate, constitution, configuration, and context controlling the Cause → Effect transition. It can add resistance, change gain, or alter the effect type.

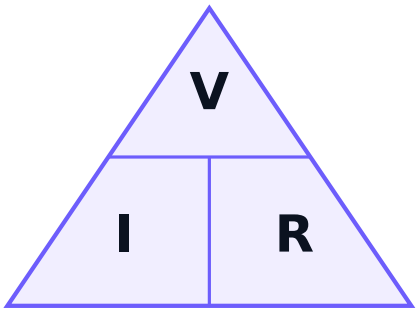
Gain role

β measures cause-to-effect amplification at constant bias. CAS-Ohm measures path flow. Do not collapse β and I into one number.

Design rule: measure the flow before the junction and the effect after the junction. A correct I reading with a wrong effect points toward bias, gain, saturation, or breakdown rather than a simple resistance problem.

WORKFLOW

The three equations and when to use them



Cover the unknown: multiply across, divide down.

A compact memory aid. Cover the unknown variable, then read the remaining operation.

1 Measure

$$I = V / R$$

Question: What attention flow happens?

Example: V=8, R=2 → I=4.

2 Design

$$V = I \times R$$

Question: What Drive is needed?

Example: target I=4, R=2 → V=8.

3 Diagnose

$$R = V / I$$

Question: What hidden constraint is implied?

Example: V=8, observed I=4 → R=2.

DEFAULT FORWARD PASS

Measure first. V and R are treated as known inputs; I is the resultant flow.

TARGET ENGINEERING

Design second. Specify a desired I, then solve for the necessary V or R.

FAULT HUNTING

Diagnose when readings disagree. Infer the constraint the system is actually experiencing.

Concise mnemonic: Measure I. Design V. Diagnose R.

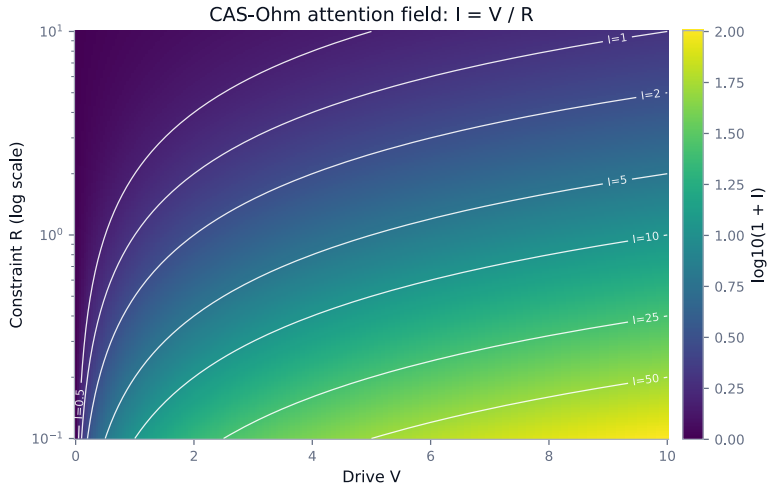
QUANTIFICATION

The decimal measurement system

0.0-10.0
DRIVE V
Zero means no applied potential. Ten is the selected board maximum.

0.1-10.0
CONSTRAINT R
Never use zero. A very small R can create very large flow.

V / R
RAW ATTENTION I
Computed. Under these bounds it can range from 0 to 100.



The attention field is steep near low resistance. That is where saturation and breakdown protection matter most.

Full-scale calibration

Choose a board-specific full-scale attention value I_{FS}. Then map raw attention to the 0-10 display scale:

```
I_Q = min(10, 10 × I_raw / I_FS)
```

Overload ratio

```
λ = I_raw / I_FS
```

λ > 1 means the raw flow exceeds the calibrated full-scale level and may push a CJT toward saturation.

Precision

Record at least one decimal place for field work and two decimals for simulation or audit trails.

Why normalize: V and R live naturally on a 0-10 design scale, while I can exceed 10. The raw value preserves engineering information; I_Q lets the four-container and card-phase systems display it consistently.

QUANTIZATION

The four-container Q4 system

Q1	Q2	Q3	Q4
0.00-2.49	2.50-4.99	5.00-7.49	7.50-10.00
latent / insufficient	emerging / partial	functional / developing	strong / mature

A proposed engineering quantizer. Each container holds a range of decimal values rather than one coarse label.

Container + decimal

Write both the exact value and its container. Example: $\mathbf{I_Q=7.82}$ **[Q4]**. The container communicates maturity at a glance; the decimal preserves precision.

Add a trend tag

Use ↑ for rising, → for stable, and ↓ for waning. Example: **V=5.40↓ [Q3]**. This prevents a mid-level state that is growing from being confused with one that is collapsing.

Q1

Insufficient, latent, blocked, or barely present.

Q2

Emerging, partial, learning, or weakly functional.

Q3

Functional and developing, but not fully mature.

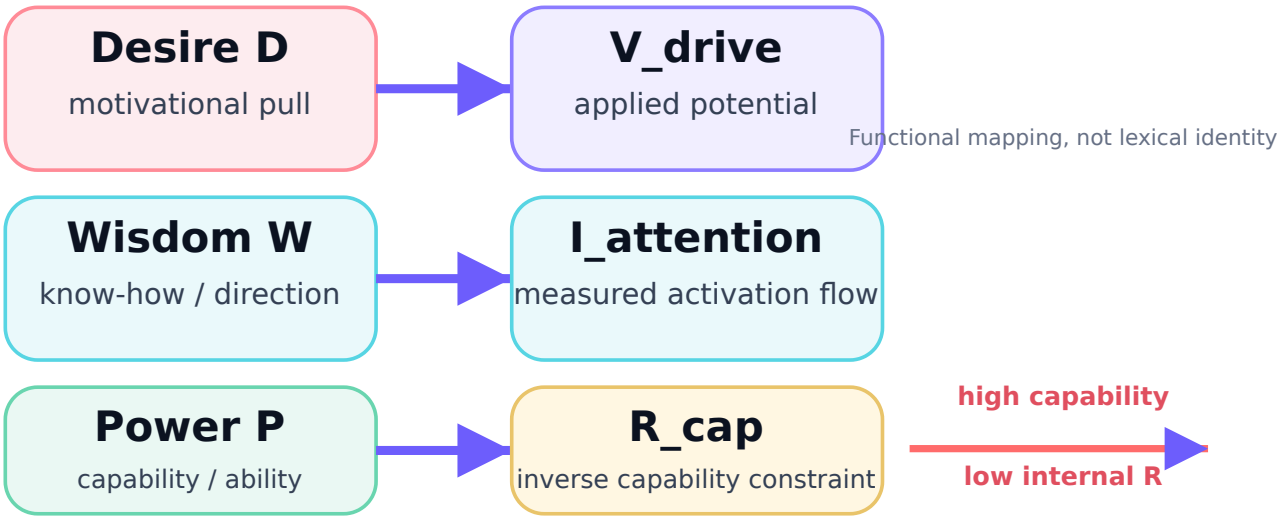
Q4

Strong, mature, or above the chosen sufficiency threshold.

Do not force the face cards into simple suit order. The canonical Desire and Power cycles include waning phases. Card phase is an anchor plus a direction, not merely a quartile.

```
Q4 notation examples
V = 8.60↑ [Q4]         # strong applied Drive
I_Q = 5.25→ [Q3]       # functional, stable attention flow
P_cap = 4.10↑ [Q2]     # developing capability
R_total = 6.20         # effective opposition on the path
```

D/W/P mapped to V/I/R, with one critical inversion



The first two links are direct functional correspondences. Power maps to the capability that sets part of R, not to resistance as a same-direction quantity.

Desire → Drive

Desire supplies motivational pressure. This is the cleanest mapping: higher Desire usually means higher available V.

Wisdom → Attention

Wisdom and attention are not synonyms. The bridge is functional: know-how guides and organizes the cognitive flow that can be observed as I.

Power → Capability

Power means ability to execute. Limited capability creates internal resistance; mature capability lowers that resistance.

External constraints remain

High Power does not erase policy, context, time, cost, law, or ethical boundaries. Those are separate resistance terms.

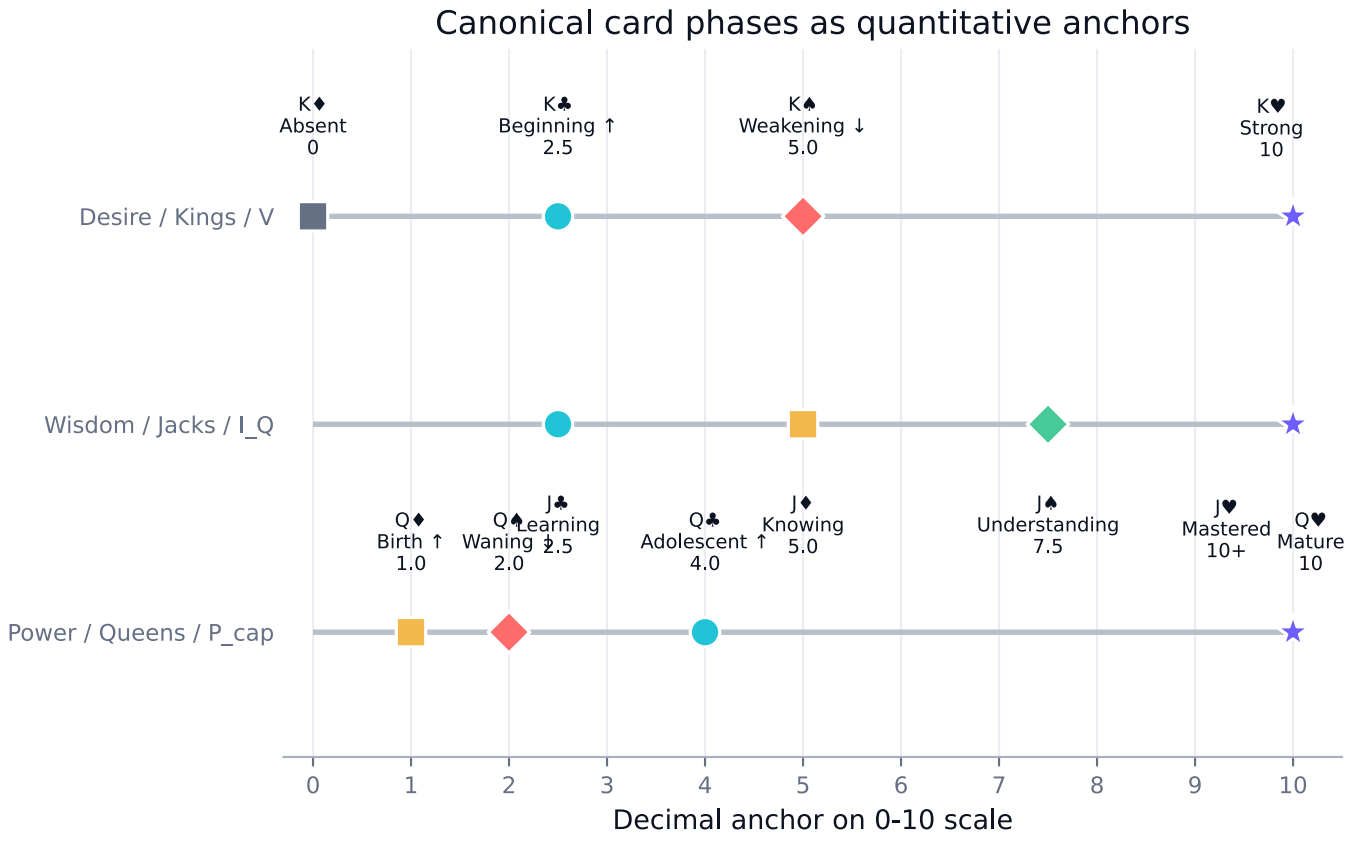
```
R_cap = R_min + (R_max - R_min) × (1 - P_cap / 10)
```

```
R_total = R_cap + R_policy + R_context + R_resource
```

Bottom line: your semantic argument is strongest when "constraint aligns with Power" means *capability-limited resistance*. The mathematical relation is inverse: more capability, less internal R.

TKS PHASE ANCHORS

The 12 prerequisite states on a decimal axis



Canonical phase percentages from the TKS v6.1 manual, expressed on a 0-10 decimal scale.

Kings: Desire / V

K♦	Absent	0.0
K♣	Beginning ↑	2.5
K♠	Weakening ↓	5.0
K♥	Strong	10.0

Jacks: Wisdom / I_Q

J♣	Learning	2.5
J♦	Knowing	5.0
J♠	Understanding	7.5
J♥	Mastered	10.0+

Queens: Power / P_{cap}

Q♦	Birth ↑	1.0
Q♠	Waning ↓	2.0
Q♣	Adolescent ↑	4.0
Q♥	Mature	10.0

Phase anchors, not rigid bins: decimals between anchors show intermediate development. The trend tag preserves whether the acquisition is rising or falling.

SOURCE TKS Formal Manual v6.1, Section 6.5. The face-card states describe how developed D/W/P is, not what D/W/P means.

INTEGRATION

A scientifically usable hybrid record

The central tension: RPM treats Desire, Wisdom, and Power as independently assessable prerequisites. Ohm's Law treats I as dependent on V and R. The solution is to use Ohm's Law as a **prediction and consistency test**, not as a replacement for the three RPM assessments.

Predictive mode

Score V and the components of R, then calculate:

$$I_{pred} = V / R_{total}$$

This is what the path should conduct if the model is complete.

Audit mode

Observe or independently rate the actual attention flow:

$$\Delta I = I_{obs} - I_{pred}$$

The residual exposes hidden resistance, amplification, clipping, or misalignment.

```
CJT-Q record
{
  cause, bias, effect,
  V_drive, P_cap, R_cap, R_policy, R_context, R_total,
  I_pred, I_obs, I_Q, delta_I,
  beta, cutoff, saturation, region,
  D_card, W_card, P_card, trend,
  test_point, timestamp
}
```

NEAR ZERO RESIDUAL

Model aligned. Predicted and observed flow agree within tolerance.

NEGATIVE RESIDUAL

Hidden opposition. More resistance or weaker bias exists than was modeled.

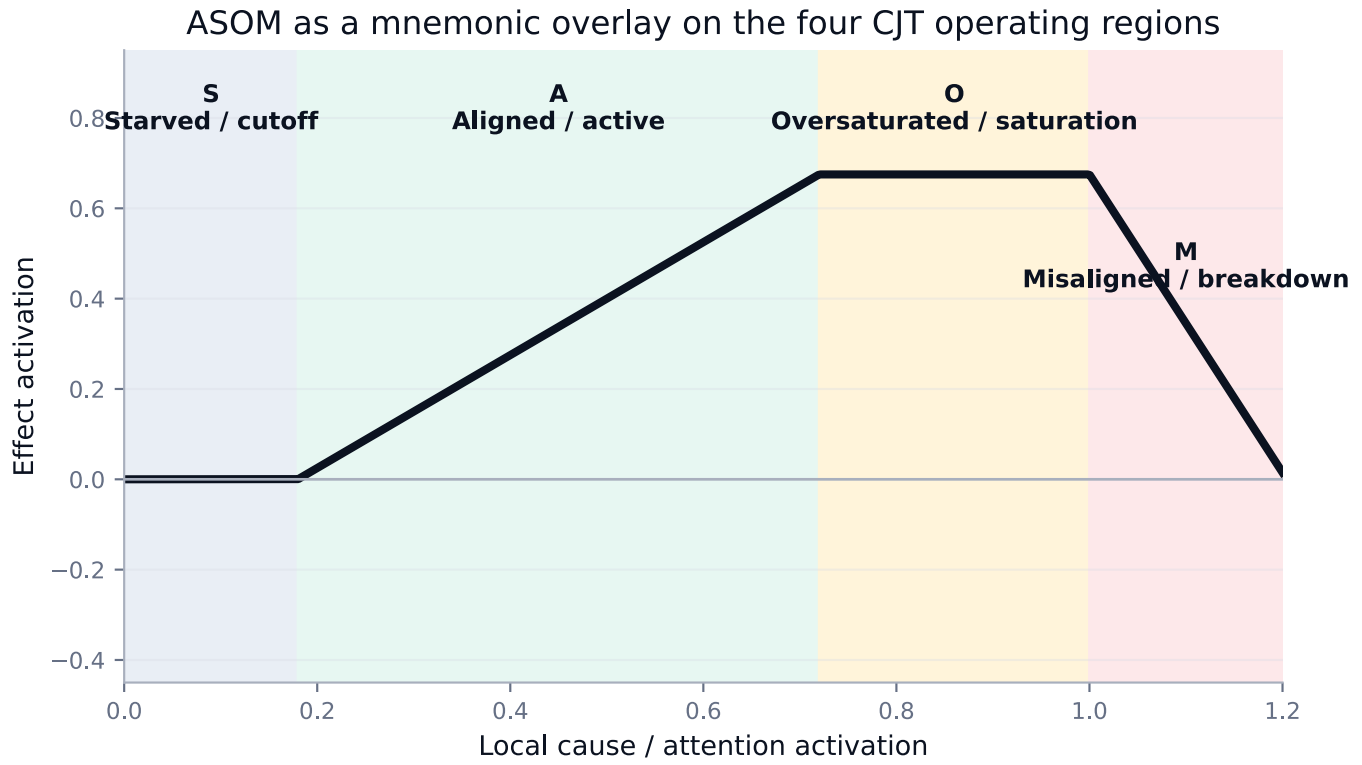
POSITIVE RESIDUAL

Extra amplification. Feedback, parallel paths, or saturation effects may be present.

```
CJTQ< cause | bias => effect > [V=8.0, R=2.0, I_pred=4.0,
                I_obs=3.7, β=1.5, region=A]
```

OPERATING POINT

ASOM mapped onto the four CJT regions



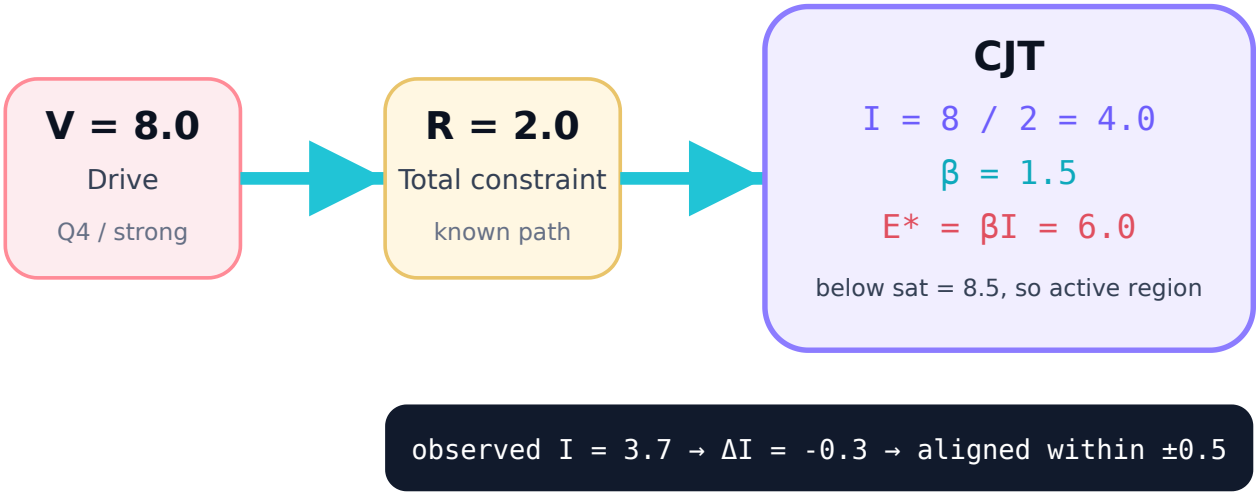
ASOM is a session-derived mnemonic. The source-defined CJT regions are cutoff, active, saturation, and breakdown.

ASOM	CJT region	Observable signature	RPM / tuning response
S: Starved	Cutoff	Bias or flow below threshold; effect near zero or too shallow.	Find the thin D/W/P leg and supply it. Add context, knowledge, time, or capacity.
A: Aligned	Active	Proportional response, small residual, correct effect type.	Confirm D/W/P coherence. Do not tune a healthy stage merely because more output is possible.
O: Oversaturated	Saturation	Effect pinned at its ceiling; more cause produces little or no additional effect.	Regulate Drive, raise deliberate constraint, bleed excess activation, or split the work across stages.
M: Misaligned	Breakdown / inversion	Effect type or direction is wrong, not merely too large or too small.	Use Wisdom to reframe, realign Desire, then apply Power last. Structural repair, not a tiny dial adjustment.

Important: a clipped or flat response can come from saturation, cutoff, or a protective branch. Classify from measured operating point and effect type, not from surface behavior alone.

WORKED EXAMPLE

One CJT, measured from rail to effect




```
observed I = 3.7 → ΔI = -0.3 → aligned within ±0.5
```

A simple example using the forward measurement equation and a separate CJT gain check. Working assumption: local I is the constrained cause activation delivered to the collector; E* = βI is therefore an effect-activation proxy, not a canonical Ohm identity.

Step-by-step

1. Apply V=8.0.
2. Model R_{total}=2.0.
3. Compute I_{pred}=4.0.
4. At β=1.5, the active-region effect proxy is 6.0.
5. Because 6.0 is below sat=8.5, the stage remains active.
6. Observed I=3.7 gives ΔI=-0.3, inside a ±0.5 tolerance.

What the example does not prove

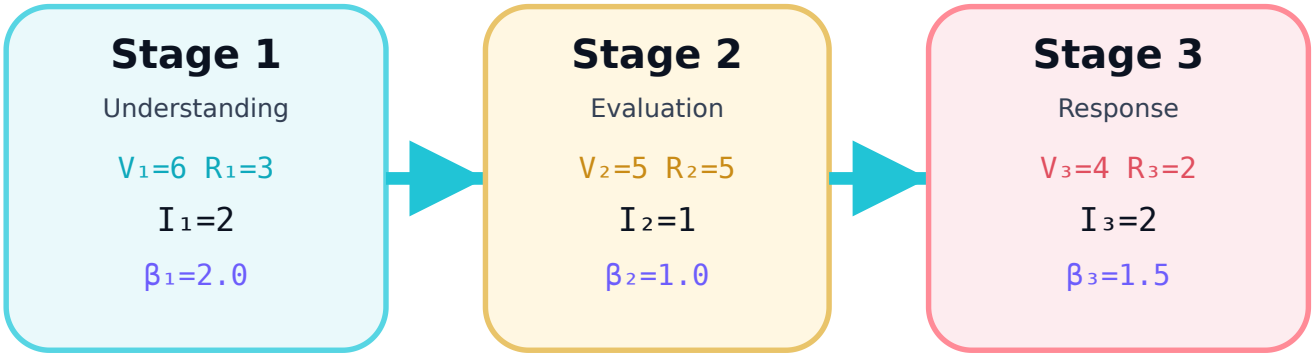
It does not prove the bias is semantically correct, the effect is the intended Noetic, or the card-phase ratings are accurate. Those require separate probes and evaluation.

Minimum probe set: input/cause activation, local I, bias state, effect activation, effect type, and region.

Engineering habit: calculate expected readings before observing the stage. A test point is most useful when you already know what a healthy reading should be.

Attention flow through a series of CJTs

Effect state from each stage becomes the next stage's cause.



$$\beta_{total} = \beta_1 \times \beta_2 \times \beta_3 = 3.0$$

Attention is measured locally at every stage.

The cascade passes effect states, not one conserved current.

Probe each stage whose failure would otherwise be silent.

A three-stage understanding → evaluation → response cascade. Each stage can have a local Drive, constraint, attention reading, bias, and gain.

What multiplies

$$\beta_{total} = \beta_1 \times \beta_2 \times \ldots \times \beta_n$$

Gain in a CJT cascade multiplies. High gain amplifies both signal and error.

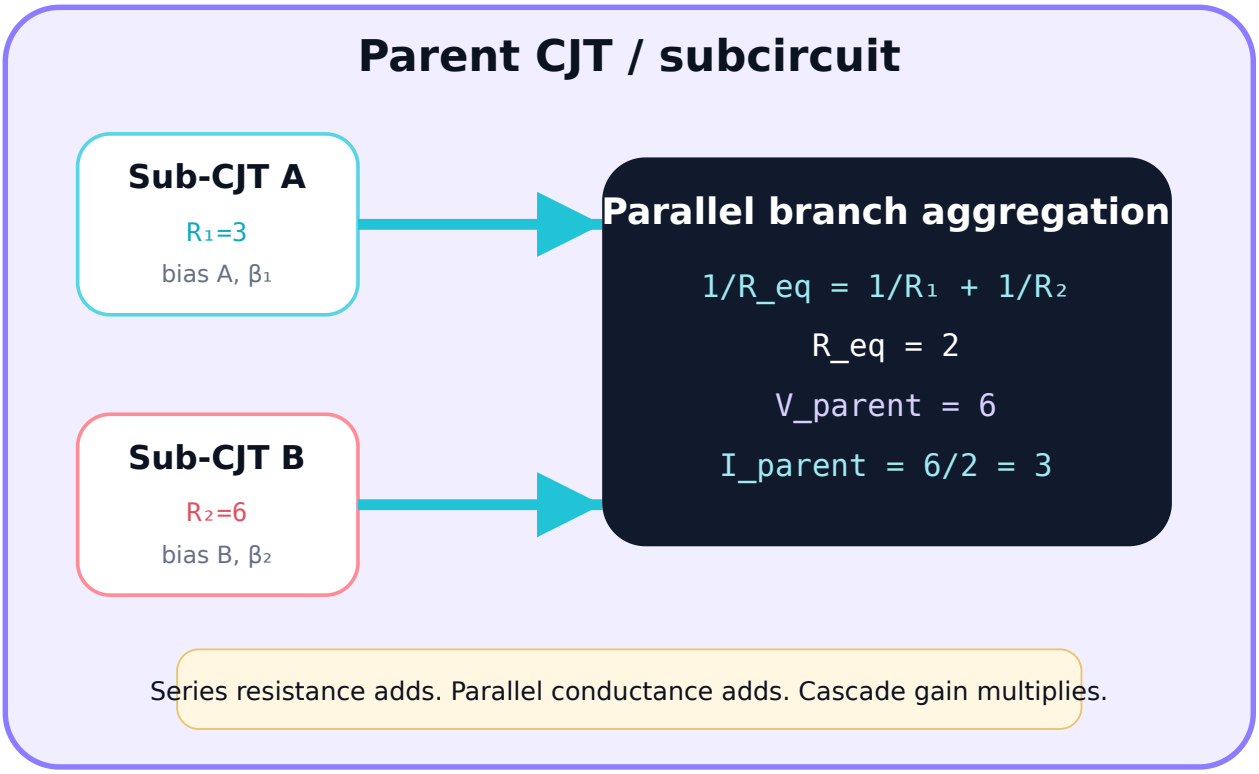
What gets handed forward

The effect state and its activation become the next stage's cause. A cognitive cascade is not identical to a wire carrying one conserved electrical current.

Common confusion: in a literal electrical series circuit, current is common through every component. In a CJT cascade, activation can be amplified, attenuated, saturated, or reshaped at each stage. Therefore read I locally at TP1, TP2, TP3, and so on.

SOURCE CAS PCBA Engineer, multi-stage amplifiers and Darlington gain: effects feed later causes and stage β values multiply.

Sub-CJTs, equivalent resistance, and parent measurement



A proposed method for reducing nested sub-CJTs to parent-level quantities while preserving child-level probes.

Series subpaths

$$R_{eq} = \Sigma R_k$$

Sequential constraints add.

Parallel subpaths

$$1/R_{eq} = \Sigma 1/R_k$$

Parallel conductances add.

Cascade gain

$$\beta_{eq} = \Pi \beta_k$$

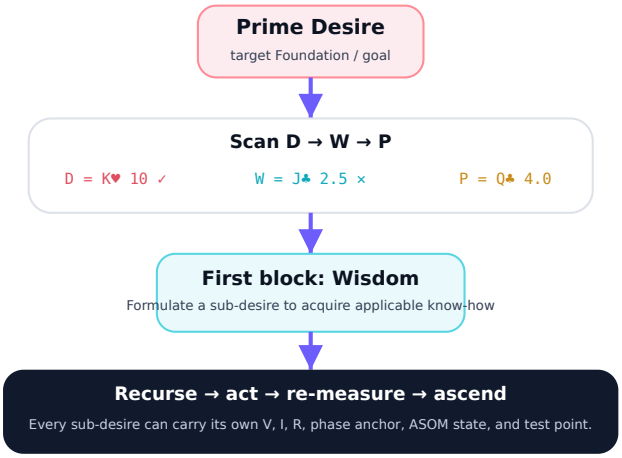
Amplification along an active series path multiplies.

Recursive measurement rule: compute the parent equivalent for overview, but never discard the child readings. A healthy parent total can hide one overloaded branch and one starved branch that cancel numerically.

```
parent {
  branch_A: V=6, R1=3  → I1=2
  branch_B: V=6, R2=6  → I2=1
  parallel: R_eq=2      → I_parent=3
  probes: TP_A, TP_B, TP_parent
}
```

RECURSIVE DIAGNOSIS

RPM as a cognitive troubleshooting circuit



RPM scans the D/W/P triad and recursively turns the first insufficient prerequisite into a sub-desire.

1. Define

Name the Prime Desire and target Foundation.

2. Scan

Assess D, W, P with card phase, decimal score, and trend.

3. Recurse

The first insufficient state becomes the next goal.

4. Act and ascend

Execute a Noetic action, re-measure, then return to the parent goal.

Canonical sufficiency anchors

Stable manifestation in the formal manual uses strong Desire, at least Understanding-level Wisdom, and mature Power as the reference condition.

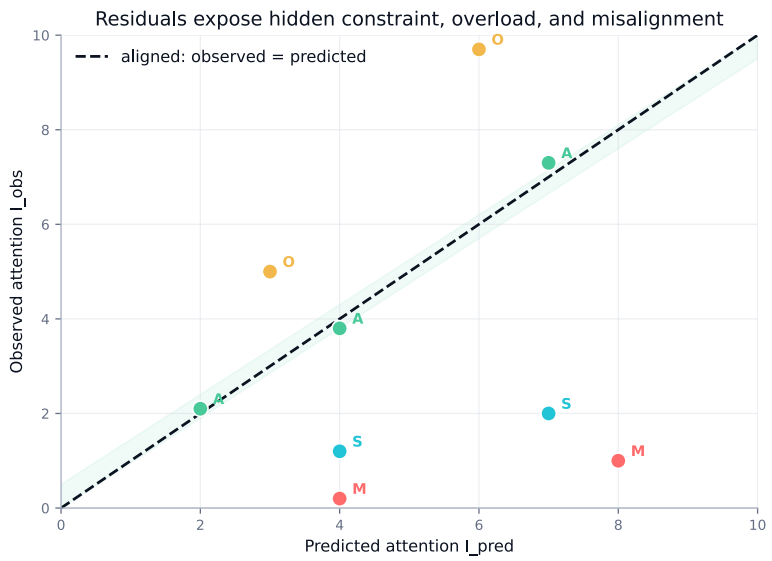
CAS-Ohm enhancement

Attach V, R, I_{pred}, I_{obs}, ΔI, and ASOM to every node. The Desire Tree becomes a quantitative fault tree.

SOURCE TKS Formal Manual v6.1, RPM Sections 7.1-7.7.

DIAGNOSTICS

Predicted versus observed attention



Illustrative classifications. The effect type and saturation state must be checked alongside the numerical residual.

Residual

$$\Delta I = I_{obs} - I_{pred}$$

Relative error

$$\varepsilon = |\Delta I| / \max(I_{pred}, \delta)$$

δ is a small floor preventing division by zero.

Hidden resistance

$$R_{hidden} = V/I_{obs} - R_{known}$$

Worked fault localization

Applied V=8 and known R=2 predict I=4. The probe reads I=2. Effective R is 8/2=4, so the path contains **2 additional units of hidden resistance**. The next probe walk asks whether that resistance comes from capability, policy, context, bias cutoff, or a miswired branch.

A

Near the diagonal and correct effect type.

S

Observed flow materially below prediction.

O

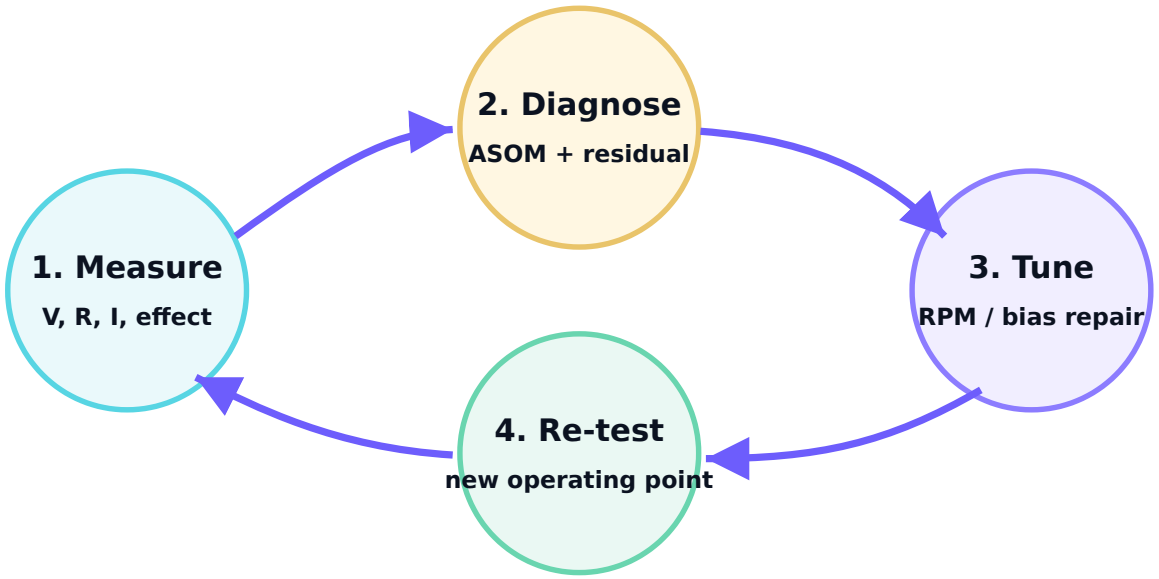
Observed/effect level exceeds linear expectation or clips.

M

Wrong direction or Noetic inversion. Residual alone is insufficient.

CLOSED LOOP

Measure, diagnose, tune, and re-test



The completed control loop. ASOM diagnoses the operating condition; RPM identifies the missing prerequisite; CAS-Ohm verifies the next operating point.

Field sequence

1. Name the target CJT, Cause, Bias, and Effect.
2. Record V and the known components of R.
3. Score Power capability and convert it to R_{cap}.
4. Compute I_{pred} and normalized I_Q.
5. Probe I_{obs}, effect activation, effect type, cutoff, and saturation.
6. Calculate ΔI and classify ASOM.
7. Run the RPM D → W → P scan.
8. Repair the first unmet prerequisite or the structural fault.
9. Re-test and preserve the before/after trace.

Worksheet

Field	Reading
Cause / Bias / Effect	_____
V / D-card / trend	_____
P_{cap} / Q-card / trend	_____
R_{cap} + policy + context	_____
I_{pred} / I_{obs} / ΔI	_____
W-card / I_Q	_____
β / cutoff / sat / region	_____
ASOM / first RPM block	_____
Repair / new reading	_____

Stop condition: do not keep tuning once the stage is aligned, active, within residual tolerance, and producing the intended effect type. More gain is not automatically better.

REFERENCE

Formula card, design rules, and source map

CAS-Ohm

$$V = I \times R$$

$$I = V / R$$

$$R = V / I$$

Measure I. Design V. Diagnose R.

Capability-derived resistance

$$R_{cap} = R_{min} + (R_{max} - R_{min})(1 - P_{cap}/10)$$

$$R_{total} = \Sigma R_{series}$$

$$1/R_{parallel} = \Sigma 1/R_k$$

Attention audit

$$I_{pred} = V/R_{total}$$

$$\Delta I = I_{obs} - I_{pred}$$

$$R_{hidden} = V/I_{obs} - R_{known}$$

CJT cascade

$$\beta_{total} = \Pi \beta_k$$

$$Effect_k \rightarrow Cause_{(k+1)}$$

Probe local I and effect at each stage.

Eight design rules

1. Never set R=0.
2. Keep raw I and normalized I_Q.
3. Preserve trend with ↑, →, or ↓.
4. Separate capability resistance from policy resistance.
5. Do not confuse Cause with Drive.
6. Do not confuse attention flow with effect.
7. Do not confuse β with CAS-Ohm current.
8. Probe child stages even when a parent aggregate looks healthy.

Source map

Topic	Primary source
CAS-Ohm, resistors, V/I/R	CAS PCBA Engineer, Ch. 5
CJT anatomy and regions	CAS PCBA Engineer, Chs. 3-4 and 9
Cascades and β product	CAS PCBA Engineer, Ch. 11
D/W/P and card phases	TKS Formal Manual v6.1, §6.5
RPM recursion	TKS Formal Manual v6.1, §7
Original prerequisite exposition	Majik and the True Kabbalah, Ch. 19

Recommended next formal step: define a versioned CAS-Ohm calibration profile for each board: V range, R components, I_{FS}, residual tolerance, saturation ceiling, and card-phase thresholds. That converts the working model into a repeatable test specification.